MINI-MOI / KEY DOCUMENT

Operations: Mini-moi

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Maintained runbook, reviewed through 2026-08-17. Use this document to identify what runs where, deploy or roll back a release, check service health, inspect schedules, find persistent data and backups, and handle credentials. Production topology and recent job outcomes were verified on 2026-08-16; release code, model routing, and German source configuration were rechecked on 2026-08-17. Design rationale belongs in `ARCHITECTURE.md`. Supporting operational evidence is in `CURRENT_STATE_OPERATIONS_2026-07-18.md`.

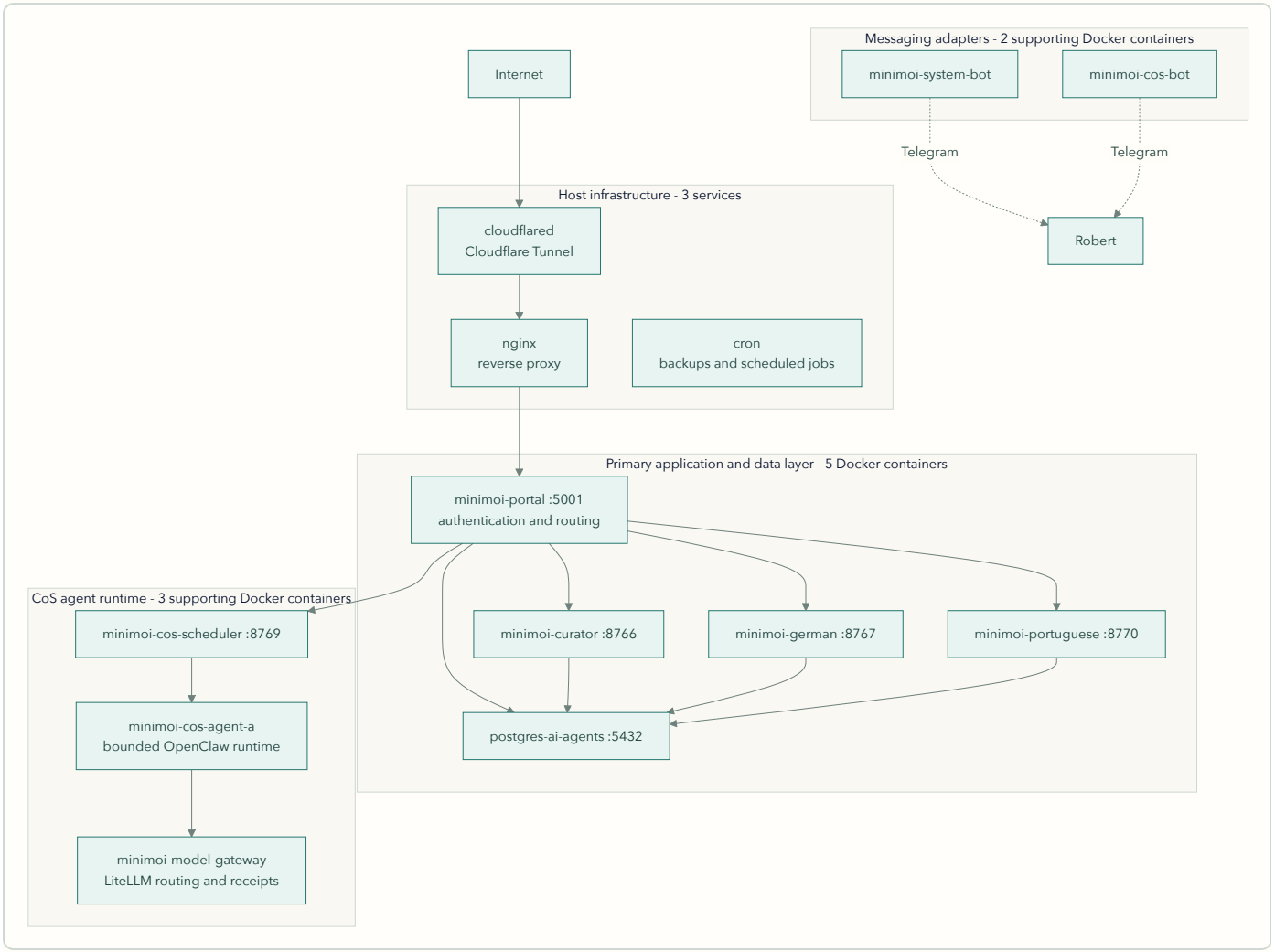
Current topology

Mini-moi runs across two nodes with distinct roles (see ARCHITECTURE.md for the design rationale; this is the operational view):

Node	Role	URL	Compose file
AWS EC2 (t3a.medium, us-east-1, `i-0d13db821169627e2`)	**Production** — live traffic, all bots, scheduled jobs	minimoi.ai	`docker-compose.prod.yml`
Mac (local, Colima)	**Dev / standby** — development, private-repo sync, DNS-switchable fallback	dev.minimoi.ai	`docker-compose.yml`

Production runtime

The primary web application is five Docker containers: the portal, three domain services, and PostgreSQL. COS Agent A adds three supporting runtime containers. Two lightweight Telegram adapters provide optional messaging channels. Three infrastructure services run directly on the EC2 host rather than in Docker.



Primary application and data layer (5 containers)

Container	Port (localhost)	Purpose
`minimoi-portal`	5001	Session authentication and routing; forwards trusted identity to the domains
`minimoi-curator`	8766	Curator Flask service
`minimoi-german`	8767	Mein Deutsch Flask service
`minimoi-portuguese`	8770	Meu Português Flask service
`postgres-ai-agents`	5432	Shared PostgreSQL data layer (`personal_agents`)

CoS agent runtime (3 supporting containers)

Container	Port (localhost)	Purpose
`minimoi-cos-scheduler`	8769	CoS scheduled-loop agent
`minimoi-cos-agent-a`	internal only	Isolated COS Agent A runtime; OpenClaw is the current swappable shell
`minimoi-model-gateway`	internal only	LiteLLM provider routing, fallback policy, health, receipts, and cost records

Messaging adapters (2 supporting containers)

Container	Port (localhost)	Purpose
`minimoi-system-bot`	none	Optional Telegram adapter for inbound system and German commands
`minimoi-cos-bot`	none	Optional Telegram chat adapter for CoS

Three things run on the host, outside Docker:

1. **nginx** — native systemd service (active since 2026-06-20), reverse proxy in front of the portal container. `systemctl status nginx` to check; config under `/etc/nginx/`.
2. **cloudflared** — the Cloudflare Tunnel process that carries public traffic to host nginx. `systemctl status cloudflared` to check it. A stopped tunnel produces Cloudflare 1033 even when the containers themselves are healthy.
3. **cron** — backup and pipeline jobs split across root and `ec2-user`. Inspect both crontabs; a container-only mental model misses this entire host layer.

All container ports bind to `127.0.0.1`. Public traffic follows Cloudflare Tunnel → host nginx → portal; no application container is directly internet-exposed.

Deploy and rollback

Deploy

Current production path: a push to `main` runs the full test/build/deploy workflow and promotes immutable commit-tagged application images. Shared and domain-specific changes currently use that same full application path.

Prepared for review, not yet live: a scoped release change exists in the isolated review worktree. After its diff is reviewed, approved, committed, and deployed, it will add three classified paths:

- **Domain-scoped release:** map changed paths to their owning services, build only those immutable images, synchronize served documents, recreate only the selected services with `--no-deps`, verify their identity and health, then prune unused images. Known coupling is explicit—for example, German also selects the system bot, while a COS change does not restart German.
- **Full application release:** shared core/topology changes and unknown paths use every application service. Ambiguity fails safely toward the full stack.

- **Document-only release:** synchronize `docs/design/`, `docs/specs/`, and the build queue through SSM. No image build, Docker Compose command, prune, or container restart occurs. The maintained root documents and generated PDFs publish in GitHub.

Release behavior and safeguards

- **Classification is fail-safe toward application deployment.** Mixed known domains use the union of their service mappings; any unrecognized runtime path uses the full path. A manual workflow dispatch is also always a full deployment. The PostgreSQL container remains host-stateful and outside the image build matrix.
- **Deployment order validates before cleanup.** The scoped script pulls the selected images, recreates only those services, verifies their immutable image tags and health, and prunes unused images only after validation succeeds. The prior container is replaced, but its image remains local during validation. If the new service fails, the script exits before pruning. This temporarily needs disk space for both releases, so check `df -h /` and `docker system df` before a large deployment. The post-success `docker image prune -af` removes unused local images; prior immutable tags remain in ECR and may need to be pulled again for rollback. The longer workflow poll and terminal-error handling remain pending until the reviewed release is shipped.
- **There is no staging gate in the deploy pipeline.** A push to `main` goes straight to production. `dev.minimoi.ai` routes via Cloudflare Tunnel to the Mac dev environment (verified in the tunnel config) — a genuinely separate origin, useful for development, but not a gate the pipeline passes through on the way to prod. A real AWS staging environment is committed work (see ROADMAP). Until then: the standing rule is that nothing is pushed without explicit approval of the reviewed diff.

Recovery options

Normal access channels: SSH (security-group allowlisted), AWS SSM Session Manager, and EC2 Instance Connect. They are independent paths for many network or credential failures, but they are not guaranteed during host exhaustion. On 2026-08-16 the instance status check failed and neither SSH nor an EC2 command-line session was usable; the AWS console reboot was the working break-glass path.

Rollback actions, typical order:

1. **`git revert` + push** — rides the same CI/CD pipeline; previous state live in ~4–5 minutes. The default for any bad deploy.
2. **Manual container recreate on the host** — can be faster than a revert when only one service is affected:

```
cd /opt/minimoi
# Verify which Compose form this host actually has before an incident, not during:
#   which docker-compose   |   docker compose version
sudo docker-compose -f docker-compose.prod.yml up -d --force-recreate <service-name>
# Use the Compose SERVICE name (e.g. curator), not the container name.
```

Production containers currently run immutable commit tags. Pulling and recreating an exact prior tag from ECR is therefore possible, but the tested command sequence still needs to be written as a formal rollback runbook.

First response and health checks (EC2)

When the site is unavailable, check the host before changing application state: EC2 status checks, memory and disk, host nginx and cloudflared, container state, then the individual health endpoints and relevant logs.

```
sudo docker ps --format "table {{.Names}}\t{{.Status}}"
df -h /                                  # 50 GiB volume; alert threshold 80%
sudo docker system df                    # image/cache headroom before deployment
free -m
curl -s http://localhost:8766/health     # curator
curl -s http://localhost:8767/health     # german
curl -s http://localhost:8770/health     # portuguese
curl -s http://localhost:5001/health     # portal
sudo docker inspect --format '{{.State.Health.Status}}' minimoi-model-gateway
sudo docker inspect --format '{{.State.Health.Status}}' minimoi-cos-agent-a
systemctl status nginx                   # host nginx — not in docker ps
systemctl status cloudflared             # public tunnel — not in docker ps
sudo docker logs minimoi-portal --tail 30 # when something looks wrong
```

Automated version: CoS's Loop H runs every 30 minutes — checks seven legacy expected containers, disk < 80%, memory < 85%, and four `/health` endpoints — alerting via the **CoS bot token**. Its expected set does not yet include `minimoi-cos-agent-a` or `minimoi-model-gateway`, and it checks neither nginx nor cloudflared. It also did not prevent the 2026-08-16 OOM incident. CoS detects and escalates within those current limits; **Robert decides the action.**

Scheduled Jobs

Host cron is split across two users and works alongside the COS scheduler container. Both crontabs and recent job logs were rechecked on 2026-08-16. Checking only one user—or only host cron—misses part of the production schedule.

User	Time (UTC)	Job	What it does
root	02:00 daily	`backup_local.sh`	Tier 1 – selected persistent paths to `/opt/minimoi/backups/YYYY-MM-DD/`
root	03:00 daily	`backup_s3.sh`	Tier 2 – S3 sync
root	04:00 Sunday	`backup_dropbox.sh`	Tier 3 – weekly Dropbox sync – **BROKEN: rclone not installed; fails silently every run**
root	5 minutes past each hour	`lesen_refresh_cli.py` in `minimoi-german`	Checks for the day's German reading refresh during the 06:00–22:00 Chicago window; retries until one run succeeds
ec2-user	hourly	`run_curator_cron_ec2.sh`	Curator pipeline – fires at most once/day, gated below
ec2-user	15 minutes past each hour	`run_intelligence_cron_ec2.sh`	AI Observations – waits for the daily briefing, runs at most once/day, and adds the weekly synthesis on Sunday; automatic daily and weekly outputs verified 2026-08-16

German's Lesen cron entry is hourly, but it is not an hourly content refresh. The command runs only from 06:00 through 21:59 Chicago time and exits after the first successful refresh of the local day. The repeated schedule supplies automatic retry when a source or network request fails. ORF Kultur and Heute are already disabled after confirmed HTTP 404 responses, so current runs do not request them. Replace or re-enable those sources only after confirming working feed URLs.

The pipeline wrapper runs hourly but only actually fires once a day, gated in order: **role guard** (exits silently unless `MINIMOI_ROLE=production` inside the container – safe on a standby node), **time gate** (12:00–20:00 UTC only), **idempotency check** (reads `briefing_date` from `curator_latest.json`; skips if today already ran). When it fires: pulls new X bookmarks (non-fatal on failure), runs the scoring pipeline, sends the Telegram briefing, stamps the date. One known wart, documented in ARCHITECTURE.md The reviewed model update makes `--model=grok-4.3` an explicit supported value; unknown values fail instead of silently selecting a default. This behavior is not live until that reviewed release is shipped.

In-container scheduling (APScheduler)

Host cron is not the whole story. `minimoi-cos-scheduler` runs its own in-process scheduler with **seven jobs** (confirmed in `domains/cos/chief_of_staff.py`):

Job	Schedule
Loop A — career focus scout	06:00 and 18:00 daily
Loop B — German watch	Sunday 09:00
Loop C — Curator watch	Sunday 10:00
Loop D — novelty watch	1st and 15th, 08:00
Loop F — build-discipline check	daily 07:30
Loop G — guest-access staleness nudge	hourly
Loop H — EC2 health check	every 30 min

The scheduler's timezone has not been explicitly confirmed — verify before relying on these clock times. Checking only `crontab -l` misses this entire layer; the full scheduling inventory is host cron (both users) **plus** this container scheduler.

Data persistence and backups

Persistence

All three domain containers mount their data directories to the host — confirmed in `docker-compose.prod.yml`. Container restarts and redeploys do not touch domain data. This closed the ephemeral-storage defect that cost the June 22–July 13 curator briefings (lost before the mounts existed — a mounts problem, historical and unrecoverable, not an ongoing risk).

Also host-mounted: `guests.json`, `build_queue.json`, `cos_memory.md`, `cos_context.json`, `docs/design/`, `docs/specs/`, and `/opt/minimoi/agent_logs/`. COS Agent A also uses named Docker volumes for runtime state and authentication; the model gateway writes receipts to `/opt/minimoi/data/model_gateway_receipts.jsonl`.

Backups: Tiers 1-2 live, Tier 3 broken

- **Tier 1 (local, 02:00 daily):** confirmed working — dated folders in `/opt/minimoi/backups/` through today. The live script captures selected Curator, interests, research-intelligence, German, Portuguese, and auth paths, then applies 14-day retention. It is not a full `/opt/minimoi/data` copy.
- **Tier 2 (S3, 03:00 daily):** confirmed working — real objects in `s3://minimoi-backups/`, logs show clean completion on consecutive recent days.
- **Tier 3 (Dropbox, 04:00 Sunday): broken, never ran successfully.** `rclone` was never installed on EC2 — the job's single log entry (its first scheduled Sunday) is `rclone: command not found`, and it has failed silently every week since. It remains scheduled and will keep failing until fixed. Tier 2 has an error trap; Tier 3 has no working failure alert.

Status corrections, both on the record: defect #136's framing of Tier 1 as "never built" was wrong — `scripts/backup_local.sh` exists and its cron runs; that half of #136 should be formally closed with a correcting comment. And the first version of *this* document claimed all three tiers verified — that verification checked cron entries and folder structure but not log output, which is exactly how a silently failing Tier 3 passed it. Verified means logs and outcomes, not schedules and folders.

Known gaps that remain true: the live backup set does not include `/opt/minimoi/cos_memory.md`, `model_gateway_receipts.jsonl`, or the `minimoi_cos-agent-a-state` and `minimoi_cos-agent-a-auth` Docker volumes. Their first production data now exists, so this is a concrete recovery gap. Tier 3 also needs `rclone` and working failure alerting. No restore test has been run; the restore test remains the planned acceptance test for staging.

Monitoring

Layer	Status	Detail
Sentry	**Live**	Wired into curator, german, portuguese servers and the portal (shipped 2026-06-23)
CoS health loop	**Live, incomplete**	Every 30 min: seven legacy containers, disk, memory, four endpoints → Telegram; excludes Agent A, model gateway, nginx, and cloudflared
CloudWatch	**Live (basic)**	EBS disk cross-check used by the health loop
Prometheus / Grafana	**Proposed, never built**	Exists only in spec documents. Do not treat any historical monitoring-stack spec as describing production.

Capacity incident, 2026-08-16. The former 2 GB `t3.small` host, with no swap, failed its EC2 instance status check after an out-of-memory event. LiteLLM was killed at roughly 631 MB and OpenClaw was using roughly 291 MB around the incident. Public traffic returned Cloudflare 1033 and SSH/EC2 command-line access failed until an AWS console reboot. Three hours after recovery, only about 146 MB remained available; the gateway used about 605 MiB and Agent A about 267 MiB. The 30 GB gp3 volume remained healthy with roughly 19 GB free. Immediate hardening is to resize memory, add a bounded swap file, set/observe container resource budgets, and add host reachability and memory alarms. The same day, production was resized to `t3a.medium` (4 GiB), its gp3 root volume was expanded from 30 to 50 GiB, and a 2 GiB swap file with swappiness 10 was added. Both EC2 checks, all ten containers, host services, and public routing passed afterward. Resource budgets and alarms remain open hardening work.

Other tracked monitoring gaps: silent Postgres failure during login `auth_id` lookup (#84), no break-glass admin account (#87), and no alert on cloudflared or Tier 3 backup failure.

Credentials and third parties

Production credentials are authoritative in AWS SSM Parameter Store under `/minimoi/production/*` and never enter git. Deployment renders the required values into the host runtime environment file for Docker Compose; that file is operational material on EC2, not a source artifact. Mac/dev uses local `.env` or Keychain-backed values and keeps them gitignored. The live parameter set is the authoritative third-party inventory:

Category	Parameters
LLM APIs	`xai_api_key`, `anthropic_api_key`, `openai_api_key`
COS agent/model gateway	`cos_agent_a_gateway_token`, `model_gateway_key`, `model_gateway_receipt_key`
Search / retrieval	`brave_api_key`, `tavily_api_key`
Translation	`deepl_api_key`
Messaging (Telegram)	`telegram_bot_token`, `telegram_cos_bot_token`, `telegram_system_bot_token`, `telegram_polling_bot_token`, `telegram_agent_bot_token`
Email	`zoho_smtp_password` (live – portal email), `gmail_app_password` (**orphaned – zero code references; resolve: delete or wire up**)
App / DB	`flask_secret_key`, `postgres_password`, `minimoi_password`, `minimoi_agent_password`, `robert_sql_password`

Telegram token → bot mapping (traced to consuming code, 2026-07-18):

SSM parameter	Consumed by	Runs where
`telegram_bot_token`	`telegram_bot.py` – outbound briefing sender	`minimoi-curator` (invoked by hourly cron)
`telegram_polling_bot_token`	`telegram_bot.py` – inline-button/callback handling	`minimoi-curator`
`telegram_system_bot_token`	`telegram_system_bot.py` – own polling loop	`minimoi-system-bot`
`telegram_cos_bot_token`	`telegram_cos_bot.py`	`minimoi-cos-bot`
`telegram_agent_bot_token`	`utils/telegram.py` – OpenClaw gateway	**Mac only** – per its own code comment

One placement note: `telegram_agent_bot_token` is scoped in code to the Mac-only OpenClaw gateway, yet it sits in the `/minimoi/production/` SSM namespace alongside the four that actually run in production. Not broken – misfiled. Cleanup candidate. One residual thread: whether `telegram_polling_bot_token` is actively exercised in production today (vs. a webhook-testing leftover) was not verified in this pass.

Curator xAI credential path: defect #152 removes Curator's dependency on an OpenClaw-managed authentication file. The scorer now uses the shared credential helper with `XAI_API_KEY`, matching German, Portuguese, CoS, the model gateway, and the deployment secret-sync step. The authoritative production parameter is `/minimoi/production/xai_api_key`; dev may use its environment or the `xai/api_key` Keychain entry. The correction reached `main` through PR #190 at `92280ea`; the successful full deployment at `72382ac` rebuilt and deployed Curator with it. The older PR #172 was closed as superseded on 2026-09-06 and must not be merged.

DB roles are separated (`robert_sql`, `minimoi_agent`) and rotated off the old weak password — confirmed as distinct SSM parameters.

COS Agent A Beta Operations

The production beta was accepted on 2026-08-16 at commit `b5bdda0`:

- both `minimoi-cos-agent-a` and `minimoi-model-gateway` reported healthy;
- an authenticated typed turn returned from COS Agent A through LiteLLM, served by the then-configured xAI `grok-4` route at fallback position 0;
- the platform-owned note path persisted a receipt-backed milestone note; and
- Robert completed a production microphone conversation and note save using the selectable OpenAI realtime voice path.

The browser never receives the long-lived Agent A or model-gateway tokens. Voice gets short-lived provider credentials and may call only the allow-listed Agent A consult and note-save tools. Production intentionally has no Ollama fallback on the current CPU host. Local Ollama remains a development/testing option. The reviewed next release replaces retired xAI aliases with explicit `grok-4.3` routing; production remains on its accepted configuration until that release ships.

Useful checks:

```
sudo docker ps --filter name=minimoi-cos-agent-a --filter name=minimoi-model-gateway
sudo docker logs minimoi-cos-agent-a --tail 50
sudo docker logs minimoi-model-gateway --tail 50
```

Do not print SecureString values while diagnosing. Confirm parameter names and versions in SSM, then run the repository's COS secret-sync script. Root console access was used for the August recovery; replacing that routine path with a least-privilege operator role remains a security follow-up.

Docs Sync

The existing `scripts/sync_docs.sh` can copy committed `docs/design/*`, `docs/specs/*`, and `data/guild/build_queue.json` to the host through SSM; root documents remain GitHub-only unless separately published. The proposed document-only CI path automates this without rebuilding or restarting containers, but that classifier/workflow is still an uncommitted review-worktree change. Do not rely on document-only deployment until its diff is approved and shipped.

Dev / Standby Environment (Mac)

The Mac runs substantially more than a mirror of the prod compose stack — this section is the fully verified picture (live `launchctl` / `docker` / `crontab` / `lsof` inspection, 2026-07-18), and most of it was previously documented nowhere.

Application layer actually running on the Mac

Service	Port	Runs as	Notes
Guild Operations agent	8768	launchd `com.user.operations`	**Guild's agents appear only here** — absent from the prod container list entirely. Code is deployed with `deploy_dev.sh deploy operations` (below)
Vault Ask	8771	launchd `com.vanstedum.minimoi-vault-ask`	`core.memory_ask.service`. The former Guild Dev agent (`com.user.devagent`) that used this port is no longer loaded (checked 2026-10-24)
CoS (bare)	8769	launchd `com.user.cos`	Runs **simultaneously** with the containerized dev CoS below — dual-instance
CoS (containerized dev)	18769	Docker `minimoi-cos-dev`	
Portuguese	8770	launchd `com.user.portuguese`	Plus scheduled `com.user.portuguese-leitura`
German	8767	launchd `com.vanstedum.german-html-server`	Plus hourly time-gated `com.vanstedum.lesen-refresh`
System bot (Mac-local standby)	—	launchd `com.vanstedum.system-bot`	**Distinct from the EC2 system-bot container** — code implements a deliberate standby/production token switch so the two never collide on the same Telegram token
cloudflared, Colima	—	launchd	Tunnel + Docker runtime
Usage report	cron 08:00 + 10:00	`scripts/track_usage_wrapper.sh`	No matching crontab entry was present on the Mac when checked 2026-07-26; restore deliberately if this report is still wanted

Dev-only by current design: Ollama (local models, `gemma3:1b`) and the nightly private-repo sync (`scripts/sync_private_repo.sh` , 02:00 local). The original personal OpenClaw remains Mac-local, while COS Agent A is a separate production container. Keychain-based credentials are the dev flow; production uses SSM.

Legacy automation still loaded: five launchd jobs from the pre-AWS era remain loaded but idle (`com.vanstedum.curator` , `curator-intelligence` , `curator-priority-feed` , `portal-boot-restart` , `minimoi-portal`). Not flagged for removal here — flagged so nobody is surprised that the Mac has more loaded automation than the active-service list suggests.

Port 8766 — resolved 2026-09-08. The earlier ambiguity is settled: the **Docker `minimoi-curator` container owns 8766**. The `ssh` process seen holding the port in `lsof` is Colima's port-forwarder acting on the container's behalf, not a native listener. Both native curator launchd jobs (`com.user.curator-server` , `com.vanstedum.minimoi-curator`) were pre-Docker leftovers that could never bind, and are disabled as `*.plist.disabled-2026-09-08` .

Deploying to dev by domain (one command per unit)

Merging to `main` is a production release, so dev is never updated by merging. Dev gets one command per **deployable unit**, a whole domain or part of one, run on the Mac from any worktree. Spec: docs/specs/spec_163_dev_deploy_by_domain_2026-10-24.md (item 163, issue #373).

```
scripts/operations/deploy_dev.sh units                                          # the
registry
scripts/operations/deploy_dev.sh deploy <unit> <commit> --operator <name> --dry-run    # checks
only
scripts/operations/deploy_dev.sh deploy <unit> <commit> --operator <name>             # build,
apply, verify
scripts/operations/deploy_dev.sh rollback <unit> --operator <name>                    # undo the
last change
scripts/operations/deploy_dev.sh status [<unit>]                                      # what each
unit runs
```

The registry is scripts/operations/dev_deploy_units.json:

Unit	Domain	Runtime	Notes
`portal`	portal	docker	Includes **the Guild UI (Build, Operate, queue) and MC chat** (`minimoi_portal/guild_ui`); moves the `board-api` sidecar with it
`guild-ui`	guild	docker, part of `portal`	Only `minimoi_portal/guild_ui` , as an overlay on the running portal image
`german` , `portuguese`	german, portuguese	docker	Deploy both for a parity fix
`cos-scheduler`	cos	docker	
`curator`	curator	docker	State stays on its host mounts; the config check refuses mount changes
`records`	records	docker	Health is the container's own healthcheck; full builds only
`operations`	guild	launchd `com.user.operations`	Guild Operations agent, 8768 (see below)
`dev-agent`	guild	launchd `com.user.devagent`	Registered, **not deployable today**: plist disabled 26 Sep 2026; 8771 is now Vault Ask
`model-gateway` , `postgres`	–	–	**Out of bounds**: refused by name

Adding a unit (a new domain or function): add one entry to `dev_deploy_units.json` (`domain` , `runtime` , `deps` , `source` , `health` ; for docker also `container` and `dockerfile` ; for launchd `label` , `script` and `state` ; `part_of` for part of a domain; spec 163 section 1 lists every field). Run `tests/test_dev_deploy.py` , which checks that the entry is complete and its files exist, then run `deploy … --dry-run` , which must show `config_diff: []` . No code change is needed.

- **What a docker deploy does:** builds the production Dockerfile over `git archive <commit>` (`--overlay` , or always for a part, copies just the unit's source onto the running image, and only when dependencies are unchanged). It labels the image with the commit and rewrites `~/minimoi-runtime/dev-pins/<service>.json` (the last file in the running compose chain). It refuses if anything but the image would change, recreates only that container, and checks it: health URLs, every deployed file (code and HTML/JS/CSS assets) against the commit, and the OpenClaw gateway untouched. For `portal` and `guild-`

`ui`, the `board-api` sidecar is stopped first, rejoined to the new portal, and proven: same namespace, same security settings, same credentials, signed readiness. A failed check rolls back by itself.

- **What a launchd deploy does** (`operations`): extracts the commit once into `~/minimoi-runtime/dev-code/<unit>/<commit>/` and links `logs/` and `data/guild/memory/` back to the agent's state home (`~/.worktrees/ops-runtime`). It re-points the plist with plistlib (script path and WorkingDirectory only), then runs `launchctl bootout` / `bootstrap`. Health means a new PID running the pinned script, `127.0.0.1:8768/health` 200 and matching source files. Rollback re-points to the previous location.
- **Record:** `~/minimoi-runtime/operators/dev-<deploy|rollback>-<unit>-<sha>-<UTC>/` (REQUEST, pin or plist before/after, config before/after as key lists and digests, health, RESULT or FAILURE/REFUSED, `rollback.sh`), plus one line per change in `~/minimoi-runtime/dev-pins/HISTORY.jsonl`.
- **It refuses** when:
  - the unit, or the domain it is part of, is held by another operator (`~/minimoi-runtime/dev-pins/HOLD-<unit>`, JSON `{"owner", "reason"}`);
  - the deploy would drop commits dev runs now (`--replace-unrelated` overrides);
  - the current code has no known commit (`--accept-unknown-current`, first deploy only);
  - migrations are involved: apply them as their own step, then `--with-migrations-applied`. Rollback never undoes schema;
  - an MC turn is in progress (portal, guild-ui);
  - another deploy or heavy job is running.
- `deploy_dev.sh` takes the host-wide lock (`_working/tools/heavy_job.py`); dry runs, `status` and `units` do not. There is no CI job for dev: reaching the Mac from GitHub would need a runner or an inbound endpoint on the host that holds the Keychain and personal data.

Mac restart protocol

Dev being down never affects production. Nothing here is urgent — but nothing here comes back on its own either, and several pieces are easy to assume are running.

1. Docker runtime, then the containers that do not self-start.

```
colima status || colima start
docker ps
```

`minimoi-cos-agent-a`, `minimoi-model-gateway`, `minimoi-curator`, `minimoi-german` and `postgres-ai-agents` carry `restart: unless-stopped` and return by themselves.

`minimoi-cos-dev` does not. Its restart policy is `no`, so it stays down after every reboot until started by hand, and a passing `docker ps` glance will not show it missing unless you know to look:

```
docker start minimoi-cos-dev
curl -s localhost:18769/health          # {"ok":true}
```

2. Verify the dev path before trusting it. Three launchd jobs carry it, and a live PID is the check that matters — a nonzero *last* exit status on a running job is normal:

```
launchctl list | grep -E 'cloudflared|minimoi-portal|com.user.cos'
curl -s -o /dev/null -w '%{http_code}\n' http://localhost:5001/        # 200
curl -s -o /dev/null -w '%{http_code}\n' https://dev.minimoi.ai/app/cos # 302 = auth redirect, healthy
```

3. Demo stacks are deliberately left stopped. The IoT Connect and Connect HQ prototypes both live in one compose file and are not needed for ordinary work. They hold roughly 275 MiB between them, which matters on an 8 GB machine.

```
cd _working/claude-connect-hq-standalone-build-2026-09-02/build/project-connect-hq
docker compose -p iotdemo start                # or: stop
docker compose -p connecthq-codex-local start  # or: stop
```

Use `stop` / `start`, never `down` — `down` removes the containers and their data. The project names are not derived from the directory, so `-p` is required.

4. `/private/tmp` is purged at boot. Disposable test virtualenvs, prepared patches and extracted evidence under `/private/tmp` do not survive a restart. This cost real time on 2026-09-19, when a reviewed test environment vanished mid-batch and looked like a tooling failure. Put anything that must outlive a reboot in the repo working tree or under `~`.

5. Low-memory triage. On the 8 GB Mac the Colima VM holds 3 GiB. If the machine is swapping hard (`vm_stat` free pages near zero, large `Swapouts`), `colima stop` returns the most memory in one move — at the cost of dev CoS, curator and German. Native Python test runs do not need it.

Known gaps and open items

Item	Tracked	Status
Curator cron wrapper masks failed scoring runs – `STATUS=$?` captures Telegram's exit code, not curator's; a failed run + successful stale send stamps `briefing_date` as success	–	**High – idempotency guarantee weaker than documented; found by Codex review 2026-07-18**
Tier 3 Dropbox backup broken – rclone never installed; fails silently weekly	–	**High – fix before the restore test; found by review 2026-07-18**
Curator xAI key depended on an OpenClaw state file	#152	Fixed and tested in review branch; production release pending
OOM recovery follow-up after resize to 4 GiB + 2 GiB swap	–	Resize complete 2026-08-16; resource budgets and memory/reachability alarms remain open
COS memory, model-gateway receipts, and Agent A state/auth volumes absent from backup set	–	**High – add to Tier 1 and verify restore before relying on them as durable memory**
Health loop omits Agent A, model gateway, nginx, and cloudflared	–	Open – extend expected services and external reachability checks
ORF Kultur and Heute Lesen feeds disabled after repeated HTTP 404	#96 follow-up	Not an active failure; replace only when working feed URLs are found
APScheduler timezone unconfirmed for the 7 in-container jobs	–	Open – verify before relying on clock times
Tier 3 backup failure does not alert	–	Open – same silent-failure family as #84
No break-glass admin account	#87	Open
Silent Postgres failure on login lookup (no logging)	#84	Open – next after current work per standing decision
No isolated AWS staging gate (Mac dev is separate but not part of promotion)	–	Acknowledged; future AWS staging
Backup restore test never run	–	Open – instrumented follow-up
`telegram_agent_bot_token` in production SSM namespace but Mac-only in code	–	Misfiled, not broken – naming/placement cleanup
`telegram_polling_bot_token` – confirm actively used in prod vs. leftover	–	Open – one residual from token mapping
Mac port 8766: native `curator_server.py` vs. Docker container both claim it	–	Open – resolve which one traffic actually reaches
Root/manual AWS access used for incident recovery	–	Replace routine production work with a least-privilege operator role; retain separately governed break-glass access
Orphaned `gmail_app_password` SSM parameter	–	Delete or wire up
#136 formally close with correcting comment	#136	Robert/OpenClaw action